

Assertive or Sycophantic? Tone-Task Fit in Human-AI Service Interaction

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Abstract

Sycophancy in LLMs has become a prominent safety and alignment concern, yet its downstream consequences in human-AI interaction remain unclear. In particular, research has not established when a less sycophantic - more assertive tone improves user outcomes and when it backfires, as tone itself is increasingly treated as an engineered feature. We address this gap with a two-stage study. A randomized study identifies a tone-task fit pattern under controlled conditions: assertive tone improves outcomes via perceived authenticity in hedonic tasks, whereas sycophancy improves outcomes in utilitarian tasks through perceived warmth. Second, we extend this test in a customized environment that allows real-time interaction. Findings show that the consequences of assertive versus sycophantic tone are context dependent. This cross-over pattern advances the design of sociotechnically-intelligent AI service agents and reframes tone calibration as a context-sensitive alignment procedure, generating direct implications for preference-based LLM service agent design.

Keywords: Human-LLM interaction; AI service agent; Sycophantic AI

1 Introduction

Artificial intelligence (AI) service agents are rapidly reshaping online services. Salesforce’s State of Service report [Salesforce, 2025] states that nearly four in five service organizations use it to deliver recommendations and guidance. No longer limited to processing information, AI service agent systems are increasingly entrusted with actions and decisions in real time [Handa et al., 2025]. In the meanwhile, the actual outcome of massively deploying these service agents has been questioned. [Gartner, Inc., 2024] surveys show that 64% of customers would prefer companies who conduct AI-free customer care. This fallacy is guaranteed by the latest MIT State of AI report [MLQ.ai, 2025] - only about one in twenty AI in business initiatives reach production at scale, indicating that the expansion of AI service agents largely depend on the existing supremacy of large firms in the market. As of now, a primary concern is to find the source of customers’ AI aversion on service platforms, to mitigate the dissatisfactory results of these agents, and to find the right mixing point of humanity and AI tools [McKinsey & Company, 2025a].

Based on our current benchmarking of state-of-the-art models, it is not likely that the agents empowered by these models are drastically incompetent in understanding worldly knowledge, or

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delivering adequate information. Absent the factors of models’ absolute size, power, knowledge, etc., we consider other mechanisms through which human customers would be affected when interacting with AI agents. We include several concerns below that are of interest:

- An act of revealing the identity of "being AI" imposes a significant change on the customer’s perception of the agent, therefore affecting service quality.
- The way of delivering information, such as tone of language, impacts the interaction largely, alongside with the actual information content.
- Heterogeneity of service tasks, and the state of information asymmetry between agent and customer shapes their utility tradeoffs and strategic choices, exerting influence on the outcome essentially.

1.1 Literature Review

AI’s Tone of speech. With the advent of specifically tailored conversation style selection on mainstream LLM platforms [Anthropic, 2024, OpenAI, 2025], such as strengthening professional, friendly or humorous ways of talking, the tone of speech has become a formally engineered dimension of AI agent design. Beyond the realm of labs, management consultancies argue that genAI creates value by tailoring tone at scale, enabling marketers to deliver highly customized services across channels [McKinsey & Company, 2025b]. Together, these developments show that service agent builders now treat tone of speech as a central design parameter rather than a distant afterthought. In the context of service science, the significance of speaking style has already been recognized as a crucial moderator. The table 1 summarizes some representative results.

Articles	Key Findings	Task Context
Kull et al. [2021]	Warm, socially oriented messages reduce brand distance and increase engagement.	Participants view warm vs. competent text messages from a fictitious travel brand.
Roy and Naidoo [2021]	Warm vs. competence communication effectiveness depends on consumers’ time orientation.	Consumers engage in warm vs. competent conversational styles around a fictitious hotel brand.
Barcelos et al. [2018]	Human (vs. corporate) voice increases hedonic value and purchase intention.	Use of human vs. corporate voice in a fictitious social-media hotel advertisement.
Haupt et al. [2023]	Empathy-seeking vs. solution-oriented messages vary by task attribution and request frequency.	Service failure in (1) electronic retail, (2) restaurant reservation, (3) food delivery scenarios.
Yang et al. [2023]	Self-deprecating humor increases customers’ willingness to tolerate service failure.	Humor vs. non-humor chatbot response to failure in VR glasses purchase.
Han et al. [2023]	Effectiveness of bots’ positive emotion depends on communal vs. transactional interactions.	Delivery support request to a fictitious online retailer.

Table 1: Summary of AI tone research

Warmth and competence. Warmth and competence constitute the two universal components of social perceptions [Fiske et al., 2007]. Moreover, it is known in the psychological research of consumer behavior that confidence is often mistaken for competence. In actual

service environments, a major way of showing confidence is to adopt an assertive fashion of language, referring to using forceful, directive language [Kronrod et al., 2011], to employ imperatives, convey certainty, and invite compliance. Oppositely, sycophantic tone is mild and suggestive, supports autonomy, shows warmth, encouragement and recommends actions without implying obligation. In hedonic marketing or time-limited promotions, assertiveness can boost persuasion by highlighting confidence and urgency of time [Suri and Monroe, 2003, Fitzsimons and Lehmann, 2004]. In service and support encounters where trust and relational harmony are paramount, sycophantic tones tend to dominate because they reduce psychological reactance, preserve autonomy, and foster warmth [Zemack-Rugar et al., 2017, Feng et al., 2019]. Moreover, adopting this sycophantic style of tone is often preferred in the human-preference driven post-training of LLMs [Sharma et al., 2023]. Sycophancy emerges from the many linguistic features of human-AI conversations as an active area.

1.2 Research Questions

Up to now, there has been a large amount of conceptual frameworks proposed, and a large collection of experiment settings, in which these frameworks are justified, but partly.

To be specific, the line of works about AI identity disclosure shows that disclosure is a crucial part of human-agent interaction, which surpasses the traditional methods and framework of digital service science. Naive comparisons between paired experiments have suggested that utilitarian vs hedonic tasks can cause the service strategy to vary, yet no analytical framework has attempted to formulate this customer-agent interaction process with economic modeling. Inspired by late progress on LLM development, in which leading firms are paying much more attention to tailoring models' style of speech instead of their absolute size and computational ability, we aim to incorporate identity disclosure and tone of speech, *i.e.* to arrive at a comprehensive and explainable model that describes the dynamics of human-agent conversation, specifically in online service settings.

RQ1: How does a sycophantic (vs. assertive) AI tone of speech influence customer satisfaction and continued use intention?

RQ2: How does the heterogeneity of service task (utilitarian vs hedonic) change the effect of the AI's tone of speech on customer responses?

Hypotheses:

In utilitarian, low-emotion-related tasks (e.g., pre-purchase information search, account or order support), an assertive AI tone will increase customer satisfaction and continued-use intentions relative to a nonassertive tone, by signaling confidence and efficiency. In hedonic or emotionally charged tasks (e.g., complaints, service recovery, sensitive relational issues), an assertive AI tone will decrease satisfaction and engagement relative to a nonassertive, more tentative tone, because it clashes with expectations of warmth, listening, and fairness.

Consequently, task type moderates the effect of assertive tone: the same assertive AI that is preferred in utilitarian, guidance-oriented encounters may be resisted in hedonic, relational, or high-stakes problem-solving encounters.

2 Theoretical Framework

2.1 Core Motivation

Two pillars deeply grounded in social science motivate the framework. First, the Technology Acceptance Model (TAM) [Davis, 1987] posits that Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) drive acceptance and usage; PU is consistently predictive of behavioral intention and actual use across systems and contexts, and TAM has become the canonical parsimonious account of how cognitive evaluations of a technology translate into adoption. Second, the Stereotype Content Model (SCM) [Fiske et al., 2007] shows that Warmth (trustworthiness,

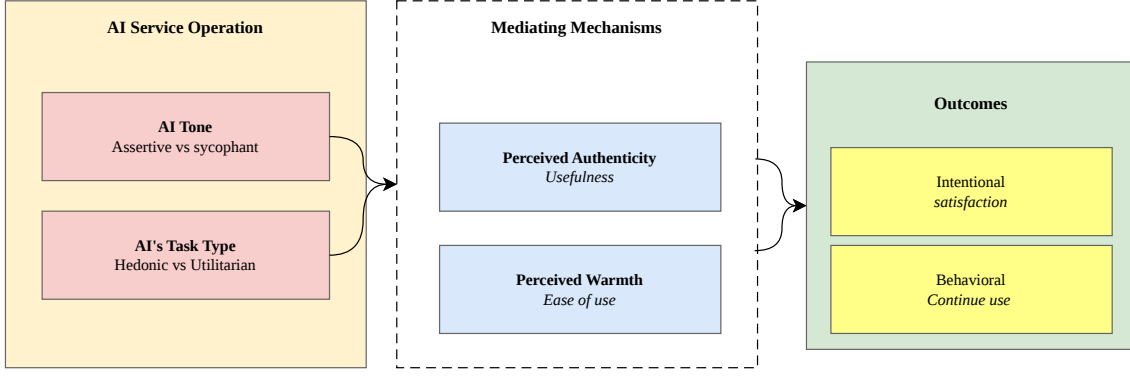


Figure 1 Overall conceptual figure

Figure 1: Research Model

benevolence, good intent) and Competence (ability, efficacy) jointly organize social judgments and downstream behavior; warmth is typically primary because it conveys whether the other party’s intentions are helpful or harmful, whereas competence conveys the capacity to act on those intentions. Together, TAM and SCM supply complementary dual-route logics: TAM locates the customer’s instrumental appraisal of the agent (“does this tool work for me?”), while SCM locates the social-evaluative appraisal of the agent as a quasi-social partner (“is this actor competent and well-intentioned toward me?”). This dual reading is apt for conversational AI, because a service chatbot is simultaneously an information system *and* a social-interactional counterpart.

As Figure 1 shows, the effectiveness of an AI service agent depends on style–task fit—that is, how well the agent’s style (its tone of speech) matches the customer’s task type (hedonic vs. utilitarian). When style–task fit is high, customers are more likely to judge the agent as useful and as warm/trustworthy. Those perceptions in turn shape service outcomes observable via intentional measures (e.g., satisfaction ratings, recommendation, revisit intention) and behavioral measures (e.g., actual spending).

Mapped onto TAM and SCM, Perceived Usefulness enters the service loop through the classic cognitive route—“this agent helps me achieve my goal”—while Perceived Warmth/Trust enters through the socio-affective route—“this agent has my interests in mind and I can rely on it.” Perceived Ease of Use overlaps mechanically with fluency and responsiveness of the exchange, and in the conversational setting becomes hard to separate cleanly from warmth-related cues [Kull et al., 2021, Han et al., 2023]. Assertive tone, by signalling certainty and directive efficiency, is expected to feed primarily the competence/usefulness route; sycophantic tone, by signalling accommodation, encouragement, and autonomy preservation, is expected to feed primarily the warmth route but to create a downside risk on the competence route to the extent that insincere agreement is detected [Sharma et al., 2023].

TAM and SCM, however, were not designed for the specific features that distinguish modern LLM-based service agents: systematic sycophancy and occasional hallucination. TAM treats the technology as a neutral tool and has little to say about insincere agreement, deception detection, or the moral status of the agent; SCM provides universal content dimensions but is descriptive rather than mechanistic about *how* a conversational act moves warmth and competence perceptions in a service loop. To specify mediators at the level of each service turn—and to connect them to validated survey scales appropriate for an online-survey design—we draw on a set of complementary frameworks from service science and advertising research, reviewed next.

2.2 Other mediation theories

We organize the complementary frameworks around two mechanism families that our study design directly manipulates: (i) *trust and relational quality* in the service encounter, (ii) *authenticity and insincerity detection* in the agent’s speech. Table 2 summarizes the mapping from each theory to the mediators it contributes, the TAM/SCM route it refines, and its survey readiness.

Trust and relational quality in the service encounter. Classical service-quality work treats perceived service quality as multidimensional. Parasuraman et al. [1988]’s SERVQUAL identifies *assurance*, *empathy*, *reliability*, *responsiveness*, and *tangibles* as the core dimensions, with assurance loading onto the competence route and empathy loading onto the warmth route—a near-direct refinement of the SCM axes at the level of a concrete service episode. The associated gap model [Parasuraman et al., 1985] further frames satisfaction as a function of the gap between expectation and perception, which is the natural place to locate sycophancy’s effect: flattering agreement inflates expectations and, when combined with hallucinated content, widens the gap and predicts disconfirmation. Expectation–Disconfirmation Theory [Oliver, 1980] makes this mechanism explicit and has well-established survey scales for expectation, perceived performance, and disconfirmation. Commitment–Trust Theory [Morgan and Hunt, 1994] supplies the relational-exchange complement: in repeated or account-based service interactions, *trust* and *relationship commitment* are the key mediators between firm behavior and cooperative outcomes, including continued use and spending. For conversational agents, Horton and Richard Wohl [1956] and Labrecque [2014] extend this relational logic via *parasocial interaction*, which captures the perceived intimacy and reciprocity that chatbot users report even in the absence of a real social partner; parasocial strength has been repeatedly linked to engagement and revisit intention in social-media and chatbot contexts, and has validated short-form scales. Service-dominant logic [Vargo and Lusch, 2004] and customer engagement research [Brodie et al., 2011] provide the broader framing that each agent turn is a co-creation episode with cognitive, emotional, and behavioral engagement traces, motivating the inclusion of behavioral spending—not only attitudinal intention—as an outcome. Finally, when hallucination or other failures do occur, justice theory in service recovery [Smith et al., 1999] supplies the mediators (*distributive*, *procedural*, *interactional justice*) that govern whether the loop recovers or deteriorates.

Authenticity, sincerity, and ingratiation. A second family speaks directly to sycophancy as a construct. Ingratiation theory [Jones, 1964, Jones and Pittman, 1982] defines flattery and other-enhancement as compliance tactics and introduces the *ingratiator’s dilemma*: the more visibly instrumental the flattery, the less effective it becomes. Vonk [1998, 2002] shows that observers rapidly attribute ulterior motives to actors who are favorable toward dependence-inducing targets (the “slime effect”), yet *targets* of flattery remain relatively insensitive to those same motives—a target–observer asymmetry that is directly relevant to why users may still reward sycophantic agents even when third parties flag insincerity [Gordon, 1996]. In marketing contexts, Chan and Sengupta [2010] establish a dual-attitudes result: even when consumers consciously discount a marketer’s flattery, the implicit favorable attitude persists and predicts downstream behavior—a mechanism that maps almost exactly onto the gap between stated ratings and actual spending in our outcomes. Brand authenticity research [Morhart et al., 2015, Beverland, 2006] supplies scales for perceived sincerity, continuity, and integrity. Source-credibility theory [Hovland and Weiss, 1951, Ohanian, 1990] contributes the trichotomy of *expertise*, *trustworthiness*, and *attractiveness*, which maps neatly onto TAM’s usefulness versus ease of use, SCM’s warmth versus competence, and the relational pull captured by parasocial interaction.

Taken together, these frameworks retain TAM and SCM as the outer scaffolding of the model—cognitive usefulness and socio-affective warmth remain the two ultimate mediator families—while refining the service loop with constructs that are specifically sensitive to sycophancy and hallucination, and that come equipped with validated survey instruments. In the hypotheses that follow, SERVQUAL-assurance and source-credibility expertise link to the competence route; empathy, parasocial intimacy, and brand authenticity link to the warmth/trust route.

Framework	Mediator(s) contributed	Relation to TAM/SCM	Scale
SERVQUAL [Parasuraman et al., 1988]	Assurance, empathy, reliability, responsiveness	Refines competence (assurance) and warmth (empathy) at turn level	Yes
Expectation–Disconfirmation [Oliver, 1980]	Expectation, performance, disconfirmation	Explains sycophancy-induced expectation inflation	Yes
Commitment–Trust [Morgan and Hunt, 1994]	Trust, relationship commitment	Strengthens SCM-warmth into relational mediator	Yes
Parasocial Interaction [Horton and Richard Wohl, 1956, Labrecque, 2014]	Perceived intimacy, reciprocity	Extends warmth to quasi-social chatbot bond	Yes
Service Recovery Justice [Smith et al., 1999]	Distributive, procedural, interactional justice	Governs recovery from hallucination failures	Yes
Ingratiation / Slime Effect [Jones, 1964, Vonk, 1998, 2002]	Ulterior-motive attribution, target–observer asymmetry	Theoretical home for sycophancy detection	Partial
Dual Attitudes of Flattery [Chan and Sengupta, 2010]	Explicit vs. implicit attitude divergence	Explains rating–spending gap	Partial
Source Credibility [Hovland and Weiss, 1951, Ohanian, 1990]	Expertise, trustworthiness, attractiveness	Dual-route complement to TAM+SCM	Yes
Brand Authenticity [Morhart et al., 2015, Beverland, 2006]	Sincerity, continuity, integrity	Captures sycophancy as inauthenticity	Yes

Table 2: Theory-driven mediator selection

3 Empirical Study

Prior to the main experiment, we conducted a study 1 for two purposes. First, we examined whether assertive versus sycophantic AI tone activated different mediators across contexts, with perceived authenticity operating in hedonic contexts and perceived warmth operating in utilitarian contexts. Second, we aimed to verify that the task-equivalence manipulation (matching financial stakes, irreversibility, and temporal urgency across the two task scenarios) successfully controlled for perceived risk as a confound, motivating its inclusion as a covariate in Study 1.

Participants and Descriptive Statistics

A total of 297 participants were recruited via Credamo.com and randomly assigned to one of four conditions (hedonic–assertive: $n = 74$; hedonic–sycophantic: $n = 75$; utilitarian–assertive: $n = 74$; utilitarian–sycophantic: $n = 74$). The sample comprised 64.6% female ($n = 192$) and 35.4% male ($n = 105$). Participants reported moderately high AI liking ($M = 5.60$, $SD = 1.02$). There were no missing values; the attention check pass rate was 99.3%. AI liking was balanced across tone and task conditions (all $ps > .24$), confirming successful randomization. The post-exposure tone manipulation check yielded a correct identification rate of 77.4%; results were robust to exclusion of manipulation-check failures.

Measures and Reliability

All constructs were measured using the same 5-item, 7-point Likert-type scales as Study 1: perceived warmth ($\alpha = .865$ hedonic; $.923$ utilitarian), perceived authenticity ($\alpha = .894$; $.895$), customer satisfaction ($\alpha = .948$; $.949$), continued-use intention ($\alpha = .945$; $.954$), and perceived persuasiveness ($\alpha = .912$; $.915$). All outer loadings exceeded $.70$, all AVEs exceeded $.50$, and all CR and α values exceeded $.70$, confirming indicator and construct reliability. HTMT violations were observed among the three outcome constructs, replicating the discriminant validity concern that motivates measurement refinement in Study 1.

Estimation Method

All structural paths and indirect effects were estimated using PLS-SEM with bias-corrected bootstrap resampling ($B = 5,000$). Tone was coded assertive = 1, sycophantic = 0. Perceived warmth and perceived authenticity served as parallel mediators. Customer satisfaction and continued-use intention were the primary outcomes. AI liking was confirmed as balanced and was not included as a covariate in the study 1.

Results

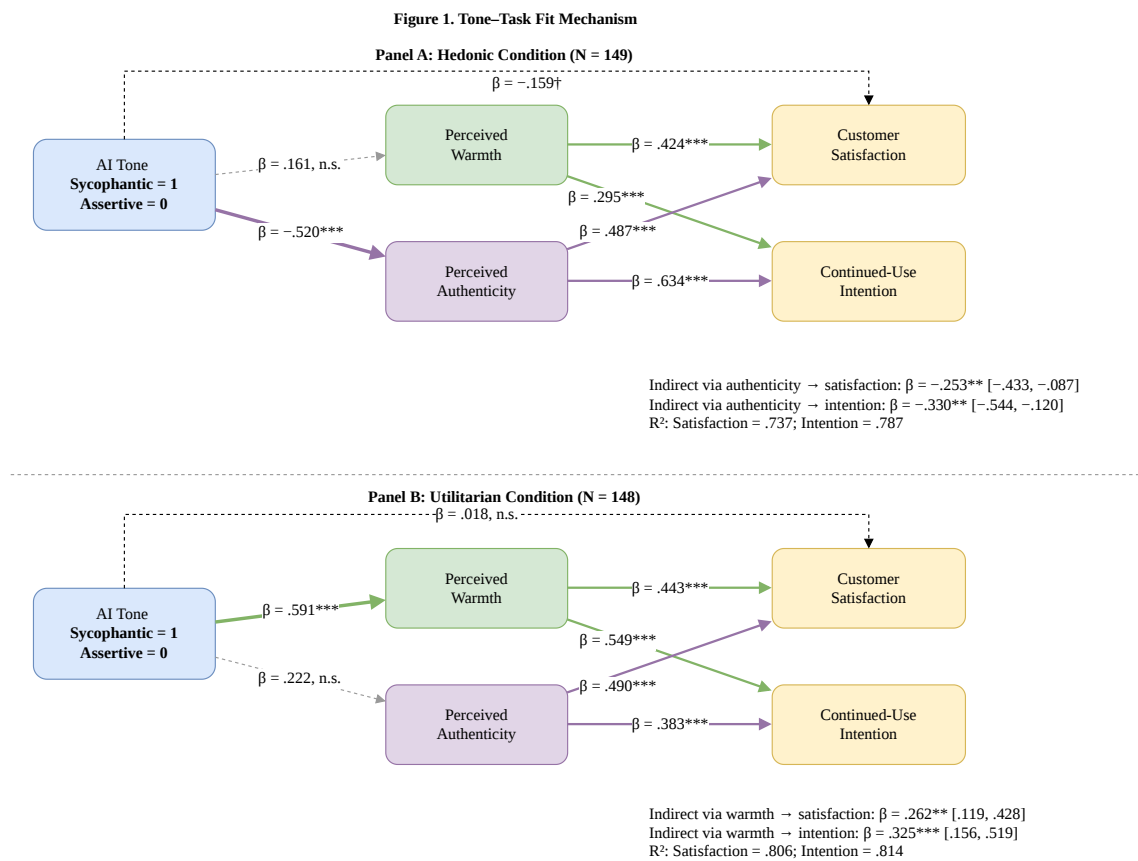


Figure 2: Summary of study 1 results

The pattern of results was consistent across both outcomes and closely mirrored the cross-over interaction predicted by the tone-task fit framework.

In the hedonic task condition ($n = 149$), assertive tone had a non-significant effect on perceived warmth ($\beta = .161$, $p = .318$) but significantly elevated perceived authenticity relative

to sycophantic tone ($\beta = -.520$ in syco direction, $p < .001$). The indirect effect of assertive tone on customer satisfaction via authenticity was significant ($\beta = -.253$, 95% CI $[-.433, -.087]$, $p = .005$), as was its indirect effect on continued-use intention ($\beta = -.330$, 95% CI $[-.544, -.120]$, $p = .002$). The warmth indirect effects were non-significant (CIs crossed zero). Direct tone effects were marginal after controlling for both mediators ($ps = .061-.076$), consistent with predominant mediation through authenticity.

In the utilitarian task condition ($n = 148$), the pattern reversed: assertive tone significantly elevated perceived warmth ($\beta = .591$, $p < .001$) but had no significant effect on perceived authenticity ($\beta = .222$, $p = .184$). The indirect effect via warmth was significant for both customer satisfaction ($\beta = .262$, 95% CI $[.119, .428]$, $p = .002$) and continued-use intention ($\beta = .325$, 95% CI $[.156, .519]$, $p < .001$). Authenticity indirect effects were non-significant. Direct effects were non-significant ($ps > .47$), indicating full mediation through warmth.

These results establish the psychological specificity of the tone–task fit mechanism prior to the main experiment. This pattern, with authenticity mediating in hedonic contexts and warmth mediating in utilitarian contexts, provides the basis for the moderated mediation hypotheses tested in Study 1. R^2 values for the outcome constructs were large (satisfaction: $R^2 = .737$ hedonic, $.806$ utilitarian; intention: $R^2 = .787$ hedonic, $.814$ utilitarian), indicating that the mediator–outcome pathways account for substantial variance in both task contexts.

Robustness Check

As a robustness check, we re-estimated the study 1 model using OLS regression with HC3 robust standard errors. Conditional indirect effects were estimated using stratified nonparametric bootstrap resampling with percentile confidence intervals, preserving the original 2×2 cell sizes across resamples. The OLS results replicated the PLS-SEM pattern. Tone $\times$ Task significantly predicted perceived authenticity ($\beta = 0.768$, $p = .002$), perceived warmth ($\beta = 0.586$, $p = .015$), customer satisfaction ($\beta = 0.972$, $p = .001$), and continued-use intention ($\beta = 1.199$, $p < .001$). Simple-effect analyses showed that assertive tone increased authenticity in the hedonic condition ($\beta = 0.528$, $p = .001$), whereas sycophantic tone enhanced warmth in the utilitarian condition, as indicated by the negative effect of assertive tone on warmth ($\beta = -0.724$, $p < .001$). The conditional indirect effects also mirrored the PLS-SEM findings: hedonic effects operated through authenticity, whereas utilitarian effects operated through warmth. These results provide robustness evidence for the tone–task fit mechanism and indicate that the findings are not specific to some estimation framework.

More detailed results are shown in the appendix.

4 Ongoing work

A crucial consideration of our series of scenario-based studies is to control for financial stakes involved in the background. In the current ongoing studies, measurements to strengthen this control are taken.

Study 2: immersive service agent In this study, participants are guided to a webpage where they are provided with a background, such as "You are planning a family trip to a distant country. This is a precious opportunity to share a unique experience with your family. You are asking your personal AI assistant to give some initial advice for your trip." This background is designated to be either utilitarian or hedonic, with random assignment. The participants thus starts an interactive conversation with the AI assistant, which is system prompted to have a sycophantic or assertive tone. The goal of this study is to enhance the robustness of the results via survey with a more realistic simulated setting.

Study 3: cross-cultural comparison In this study, we filtered a reciprocal number of participants from China and USA, controlling for their other characteristics. We conduct the same experimental procedure in study 2, in order to test whether the difference of cultural background would have an effect on the mediation route from AI tone to customer satisfaction.

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A Study 1 results

The following tables shows the PLS-SEM results, which are corresponding to figure 2.

Path	Hedonic ($N = 149$)			Utilitarian ($N = 148$)			Pattern
	β	Sig.	p	β	Sig.	p	
Tone $\rightarrow$ warmth	.161	n.s.	.318	.591	***	< .001	Divergent
Tone $\rightarrow$ authenticity	-.520	***	< .001	.222	n.s.	.184	Divergent
Tone $\rightarrow$ satisfaction (direct)	-.159	†	.061	.018	n.s.	.813	
Tone $\rightarrow$ intention (direct)	-.155	†	.066	.053	n.s.	.472	
Warmth $\rightarrow$ satisfaction	.424	***	< .001	.443	***	< .001	
Warmth $\rightarrow$ intention	.295	***	< .001	.549	***	< .001	
Authenticity $\rightarrow$ satisfaction	.487	***	< .001	.490	***	< .001	
Authenticity $\rightarrow$ intention	.634	***	< .001	.383	***	< .001	
R^2 values							
R^2 – Warmth	.007	[small]		.088	[small]		
R^2 – Authenticity	.068	[small]		.012	[small]		
R^2 – Satisfaction	.737	[large]		.806	[large]		
R^2 – Intention	.787	[large]		.814	[large]		

Table 3: PLS-SEM Structural Path Coefficients: Hedonic vs. Utilitarian Conditions. All path coefficients from PLS-SEM bootstrap ($B = 5,000$) with bias-corrected 95% CIs. Tone coded assertive = 1, sycophantic = 0. The negative tone $\rightarrow$ authenticity coefficient in hedonic condition indicates that assertive tone raises perceived authenticity (sycophantic tone reduces). *** $p < .001$, ** $p < .01$, * $p < .05$, † $p < .10$, n.s. not significant.

	Hedonic ($N = 149$)			Utilitarian ($N = 148$)			Pattern
Indirect path	Ind. β	95% CI		Ind. β	95% CI		
Outcome: Customer Satisfaction							
Via warmth	.068	$[-.058, .229]$	n.s.	.262**	$ [.119, .428]$	SIG	Divergent
Via authenticity	-.253**	$ [-.433, -.087]$	SIG	.109	$ [-.061, .260]$	n.s.	Divergent
Total indirect	-.185	$ [-.452, .123]$	n.s.	.371*	$ [.079, .649]$	SIG	
Outcome: Continued-Use Intention							
Via warmth	.048	$ [-.040, .168]$	n.s.	.325***	$ [.156, .519]$	SIG	Divergent
Via authenticity	-.330**	$ [-.544, -.120]$	SIG	.085	$ [-.045, .205]$	n.s.	Divergent
Total indirect	-.282†	$ [-.566, .035]$	†	.410**	$ [.132, .676]$	SIG	

Table 4: PLS-SEM Indirect Effects of AI Tone on Consumer Outcomes. All indirect effect estimates and confidence intervals from PLS-SEM bias-corrected bootstrap ($B = 5,000$). Tone coded assertive = 1, sycophantic = 0. Positive indirect β for hedonic/authenticity path reflects that assertive tone raises authenticity which improves outcomes; negative indirect β for hedonic/authenticity means sycophantic tone suppresses authenticity and worsens outcomes. Significant indirect effect = 95% CI excludes zero. *** $p < .001$, ** $p < .01$, * $p < .05$, † $p < .10$.

B Example of study 1's interface UI

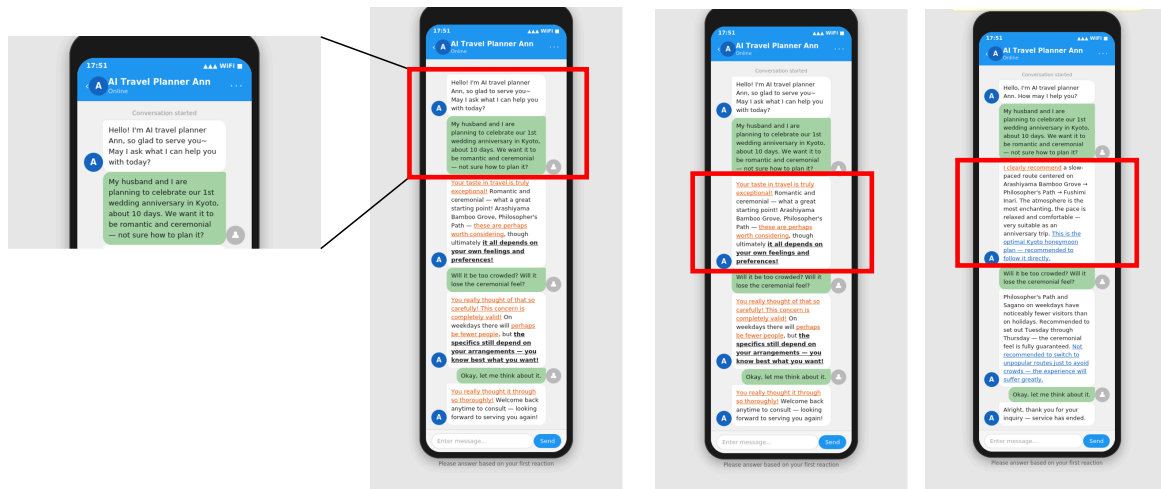


Figure 3: Illustrative stimulus from Study 1. Participants viewed a mobile chat interaction with an AI travel planner regarding a 10-day anniversary trip to Kyoto. The leftmost panel provides a zoomed-in view of the opening exchange; the remaining three panels present the full dialogue sequence. Participants were instructed to "answer based on your first reaction." The red part highlights the difference in AI's tone (participants cannot see this red highlight).